

EP 07 The Coefficient of Viscosity

1. Objective

To familiarize with the use of a micrometer and a vernier caliper and measure the coefficient of viscosity of castor oil.

2. Principle

Viscosity is an internal friction property exhibited by all fluids. Liquids like castor oil, which flow slowly, have higher viscosity than water. This experiment relies on **Stokes' Law** to calculate the viscosity coefficient (η).

2.1 Stokes' Law

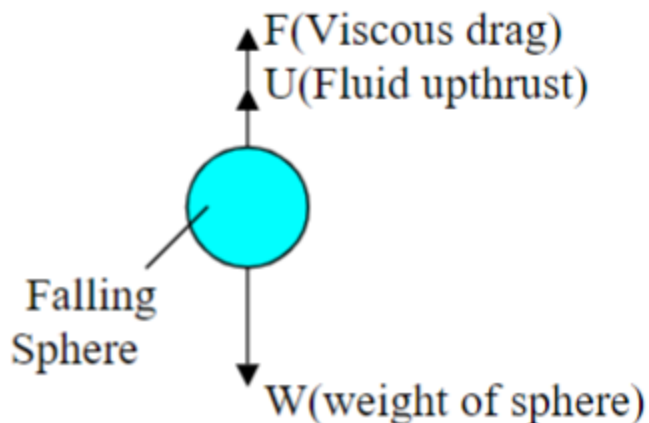
The viscous drag force F acting on a sphere moving steadily in an infinite fluid is given by: $F = 6\pi\eta vr$, where η is viscosity coefficient, v is velocity of the sphere, and r is radius of the sphere. This law is only valid for steady motion in an *infinitely extended fluid*; vessel walls/bottoms distort the drag force if the fluid is confined.

2.2 Force Balance and Terminal Velocity

When a sphere falls vertically in viscous castor oil, three forces act on it

- Downward weight: $W = \frac{3}{4}\rho\pi r^3 g$ (ρ is density of the sphere)
- Upward buoyancy: $U = \frac{3}{4}\sigma\pi r^3 g$ (σ is density of castor oil)
- Upward viscous drag: $F = 6\pi\eta r v_t$

The sphere accelerates until the resultant force is zero, reaching a constant **terminal velocity** v_t . At equilibrium $W - U - F = 0$.



Substituting W , U , and F into the equilibrium equation and rearranging gives:

$$v_t = \frac{2r^2(\rho - \sigma)g}{9\eta}$$

2.3 Calculation of Viscosity Coefficient

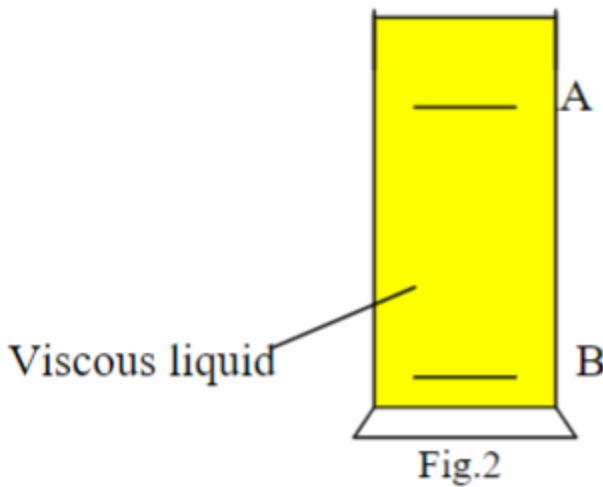
Terminal velocity is measured by

$$v_t = \frac{L}{t}$$

, where L is distance between marks A and B, and t is average time for the sphere to fall from A to B (A is far below the surface to ensure v_t is reached; B is not near the bottom). Then we can obtain the equation of η :

$$\eta = \frac{(\rho - \sigma)g}{18L} d^2 t$$

where $d = 2r$, r is the radius of the ball and d is the diameter of the ball.



Since the fluid is confined, vessel walls affect motion. The corrected formula accounts for this:

$$\eta = \frac{(\rho - \sigma)g}{18L} d^2 t \left(1 - 2.4 \frac{d}{D} - 3.3 \frac{d}{2h} \right)$$

where D is the inner diameter of the vessel, and h is the height of castor oil .

3. Procedure

1. **Measure Basic Parameters:** Record the initial (T_{begin}) and final (T_{finish}) temperatures of castor oil to calculate the average temperature ($T_{average}$). Measure the density of the sphere (ρ_{ball}) and castor oil (σ_{oil}), distance L between A and B, vessel inner diameter D , and oil height h . Complete table 1.

- Measure Sphere Diameter:** Use a micrometer to measure the diameter (d_1) of Ball 1 three times.
- Measure Falling Time:** Release Ball 1 from the top of the oil. Record the time (t_1) it takes to fall from mark A to B, repeating three times.
- Repeat for Second Sphere:** Repeat steps 2–3 for Ball 2 (measuring d_2 and t_2 three times each). Complete table 2.

4. Data Recording and Processing

4.1 Data Form

Table 1: Basic Parameters

$T_{begin}=$	$T_{finish}=$	$T_{average}=$
$\rho_{ball}= \quad \quad \quad g/cm^3$	$\sigma_{oil}= \quad \quad \quad g/cm^3$	$L= \quad \quad \quad cm$
$g= \quad \quad \quad cm/s^2$	$D= \quad \quad \quad cm$	$h= \quad \quad \quad cm$

Table 2: Sphere Measurement Data

item times	Ball 1			Ball 2		
	$d_1(cm)$	$t_1(s)$	$\eta_1(P)$	$d_2(cm)$	$t_2(s)$	$\eta_2(P)$
1						
2						
3						
Average						

4.2 Data Calculation

- Reference value: $\eta_{reference} = 6.20P$
- Viscosity for Ball 1: η_1
- Viscosity for Ball 2: η_2
- Average of Viscosity: η_{avg}
- Uncertainty Estimation: U_η
- Final Result: $\bar{\eta}$

4.3 Data Analysis

- Data Validity Check:
- Impact of Vessel Correction Term:
- Result Consistency:

5. Error Analysis

5.1 Systematic Errors

- **Vessel Wall Effect:** Even with the correction formula, residual errors from the finite vessel cannot be fully eliminated, as the correction is empirical.
- **Instrument Precision:** The micrometer (minimum division: 0.01 mm) and stopwatch (0.1 s resolution) have inherent measurement limits, leading to biases in d and t .
- **Temperature Variation:** Viscosity is highly temperature-dependent (castor oil viscosity decreases with increasing temperature). Small temperature fluctuations cause errors in η .

5.2 Random Errors

- **Human Reaction Time:** Delays in starting/stopping the stopwatch lead to variations in t measurements.
- **Sphere Uniformity:** Imperfectly spherical balls cause inconsistent drag forces, affecting falling time repeatability.

6. Questions

1. Why is a correction term added to the viscosity formula?

The correction term accounts for the effect of the finite vessel walls and bottom, which increase viscous drag beyond the prediction of Stokes' Law (for infinite fluids).

2. How does temperature affect the experiment result?

Viscosity of castor oil decreases significantly with increasing temperature. Higher temperatures reduce oil internal friction, decreasing the falling time (t) and thus the calculated η .